A REPORT

ON

APEX PORTAL :CONSTRUCTION SAFETY

Submitted by

Mr.CHETHAN G LOKKLLER - 20241CSE0577

Under the guidance of,

Mr. Vishnu Shankar

Ms. Akkamaha Devi

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

AT

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

PRESIDENCY UNIVERSITY

BENGALURU

AUGUST 2026

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on **APEX PORTAL** being submitted by **CHETHAN G LOKKLLER** bearing roll number **20241CSE0577** in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering** is a Bonafide work carried out under our supervision.

Mr. Vishnu Shankar
Assistant Professor
PSCS
Presidency University

Ms. Akkamaha Devi
Assistant Professor
PSCS
Presidency University

Ms. Vidhya Rengasamy
Internship coordinator
PSCS
Presidency University

Dr. Bhuvaneshwari Patil
School level Internship coordinator
PSCS
Presidency University

Dr. Nagaraja S R
Head of the Department
PSCS
Presidency University

Dr. Duraipandian N
Dean – PSCS & PSIS
Presidency University

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I hereby declare that the work, which is being presented in the report entitled "**APEX PORTAL:AI CONSTRUCTION SAFETY:"** in partial fulfillment for the award of Degree of **Bachelor of Technology** in **Computer Science and Engineering ,** is a record of my own investigations carried under the guidance of **Mr. Vishnu Shankar, Assistant Professor, PSCS, Presidency University and Ms. Akkamaha Devi, Assistant Professor, PSCS, Presidency University,** Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

Name: CHETHAN G LOKKLLER
Roll No.: 20241CSE0577
Signature of the Student:

INTERNSHIP COMPLETION CERTIFICATE

- **Add Internship/Course completion certificate issued from an organization. It must have the duration of the Internship, i.e. start and end date, Internship/RPL/Skill based course title and technology on which work is carried out.**

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Name: CHETHAN G LOKKLLER

Roll No:20241CSE0577

ABSTRACT

APEX PORTAL – Construction Safety is an intelligent, AI-powered construction-site safety monitoring and analysis platform developed to improve workplace safety by automating the identification of Personal Protective Equipment (PPE) violations and supporting continuous visual monitoring. Construction sites are dynamic environments where workers operate around machinery, construction materials, elevated structures, vehicles, and other potential hazards. In such environments, ensuring that every worker consistently follows required safety procedures is a challenging task. Conventional safety monitoring methods mainly depend on manual inspection and human supervision. Although these methods are important, they can be time-consuming, difficult to scale, and limited in their ability to continuously observe all workers and locations. The proposed APEX PORTAL addresses these challenges by integrating Artificial Intelligence, Computer Vision, Deep Learning, image processing, and data visualization into a unified safety monitoring platform.

The primary objective of the project is to develop an automated system capable of analyzing construction-site images and video streams to identify workers and verify the presence of essential PPE, particularly safety helmets and safety vests. The platform utilizes the YOLOv11 object detection model from Ultralytics as its core computer vision model. YOLOv11 enables the system to locate and classify relevant objects within individual images and video frames by generating bounding boxes and confidence scores. Based on the detected objects, the system can evaluate PPE compliance and identify situations where required safety equipment is missing. This approach enables the transformation of raw visual information into meaningful safety-related information that can be reviewed by supervisors and safety personnel.

The development process begins with the preparation of a suitable construction-safety dataset. Roboflow is used for dataset collection, annotation, organization, preprocessing, and management. The prepared dataset is divided into appropriate training, validation, and testing sets to support model development and evaluation. The YOLOv11 model is trained and tested using Google Colab, with NVIDIA GPU acceleration used to improve computational performance during deep-learning training. The trained model is subsequently integrated into the application for performing object detection and PPE compliance analysis on construction-site images and video data.

Python is used as the primary programming language for implementing the overall application and connecting the different components of the system. OpenCV is used for image and video processing, frame extraction, resizing, image manipulation, and visualization of detection results. Pandas is used to organize detection information, maintain safety records, process collected data, and support the preparation of safety reports. Matplotlib is used to generate graphical representations of safety-related information, enabling users to understand detection trends and violation statistics more

easily. Streamlit is used to develop an interactive web-based dashboard that provides a centralized interface for monitoring and analyzing construction-site safety information.

The system follows a structured processing pipeline. First, construction-site images or video streams are provided as input. The input data is processed using OpenCV, and video streams are divided into individual frames when required. Each frame is passed to the trained YOLOv11 model for object detection. The model identifies workers and relevant PPE objects and produces detection results with corresponding labels and confidence values. The system then evaluates the relationship between detected workers and required safety equipment to determine PPE compliance. When a worker is identified without the required helmet or safety vest, the platform records the corresponding safety violation. The detected information can then be displayed through the dashboard and used for generating safety analytics and reports.

APEX PORTAL is designed to provide a clear and centralized view of construction-site safety conditions. The dashboard can display important information such as the number of workers detected, PPE compliance status, helmet violations, safety-vest violations, total safety violations, compliance percentages, safety scores, and other relevant indicators. Graphs and visualizations can be used to represent safety trends and violation frequencies, allowing safety personnel to identify recurring problems and understand the overall safety performance of the monitored environment. The system can also support alert mechanisms to highlight detected violations and draw attention to situations that may require further inspection or intervention.

One of the important aspects of the project is the conversion of computer vision outputs into useful safety intelligence. Instead of simply detecting objects, the platform organizes detection results into structured safety information. This information can be used to compare compliant and non-compliant observations, identify frequently occurring PPE violations, monitor safety performance over time, and support safety-related decision-making. Such an approach can help organizations move from purely reactive safety inspection toward a more proactive and data-driven monitoring process.

The platform also provides a foundation for future expansion. Additional PPE classes such as safety gloves, goggles, safety shoes, and other protective equipment can be incorporated by extending the dataset and retraining the detection model. Future versions can also include advanced computer vision capabilities such as restricted-zone monitoring, worker fall detection, fire and smoke detection, unsafe proximity to machinery, crowd monitoring, and multi-camera surveillance. Integration with cloud platforms, mobile notifications, IoT devices, and centralized safety management systems could further improve the usefulness and scalability of the solution.

The key contribution of APEX PORTAL – Construction Safety is the integration of AI-based object detection, automated PPE compliance analysis, video processing, safety data management, and interactive visualization into a single platform. The system reduces the need for repetitive manual observation and provides a faster method for identifying visible safety violations. While the platform is intended to support safety personnel rather than replace professional safety procedures and human judgment, it can serve as an additional intelligent monitoring layer that helps bring important safety observations to attention more efficiently.

Overall, APEX PORTAL – Construction Safety demonstrates how modern Artificial Intelligence, Computer Vision, and Deep Learning technologies can be applied to address real-world construction safety challenges. By automatically analyzing visual information, detecting PPE-related violations, organizing safety data, and presenting actionable analytics through an interactive dashboard, the platform provides a practical foundation for intelligent safety management. The proposed system contributes toward creating safer, more efficient, automated, and data-driven construction environments while providing opportunities for further development into a comprehensive AI-based construction safety intelligence solution.

INTRODUCTION

The construction industry is one of the most important sectors contributing to infrastructure and economic development. However, construction sites are also environments where workers are exposed to various **physical hazards, unsafe activities, heavy machinery, falling objects, and environmental risks**. Therefore, maintaining proper safety practices and ensuring that workers use appropriate **Personal Protective Equipment (PPE)** is an essential part of construction-site management.

Common PPE such as **safety helmets, safety vests, protective footwear, gloves, and other protective equipment** helps reduce the risk associated with workplace hazards. Despite the availability of safety guidelines, ensuring that every worker consistently follows these requirements can be difficult. Traditional safety inspection generally depends on supervisors and security personnel manually observing workers, which can be challenging when a site contains a large number of workers and activities are taking place simultaneously.

Manual monitoring can also become difficult for **continuous and large-scale surveillance**. Safety personnel may not be able to observe every worker at every moment, and violations may sometimes remain unnoticed until an inspection is performed. This creates a need for technologies that can assist safety teams by providing **automated and continuous monitoring** of construction activities.

Recent developments in **Artificial Intelligence (AI), Computer Vision, Machine Learning, and Deep Learning** have created new possibilities for automated safety management. Computer Vision systems can analyze images and video streams and extract useful information about objects, workers, and their surroundings. Object detection models can further identify specific objects within an image and determine their locations with a high level of automation.

APEX PORTAL – Construction Safety is developed as an AI-powered platform that applies these technologies to the construction safety domain. The system uses the **YOLOv11 object detection model** to identify workers and monitor PPE compliance. It focuses on detecting safety equipment such as **helmets and safety vests** and identifying situations where required PPE is not properly used.

The platform processes construction-site images and videos and uses computer vision techniques to identify relevant objects and safety conditions. When a PPE violation is detected, the system records the information for further analysis. This allows safety personnel to obtain a clearer understanding of the safety conditions at the site without depending entirely on manual observation.

The project also incorporates **Roboflow** for dataset preparation and management, **Google Colab** for model training, and **NVIDIA GPU acceleration** to improve the efficiency of AI model development. **Python and OpenCV** are used for image and video processing, while **Pandas and Matplotlib** support data handling and visualization.

To make the collected information easier to understand, **APEX PORTAL** includes an interactive **Streamlit-based safety dashboard**. The dashboard can present important information such as **PPE violations, compliance levels, safety scores, risk levels, violation trends, and safety records**. This provides a centralized platform for reviewing and analyzing construction-site safety information.

The system can also support **AI-powered video analysis**, allowing construction-site footage to be examined for worker and PPE-related safety conditions. The information generated from the detection process can be used to support faster identification of safety violations and assist safety teams in taking appropriate action.

Another important aspect of the project is **data-driven safety analysis**. Instead of simply detecting an individual violation, the platform can organize detection results into meaningful safety information. This can help users understand the overall level of PPE compliance, identify recurring violations, and monitor changes in safety performance over time.

The main purpose of **APEX PORTAL – Construction Safety** is not to completely replace human safety professionals but to provide them with an **AI-assisted monitoring and decision-support tool**. By combining automated detection, safety analytics, and dashboard-based visualization, the system can help reduce the workload associated with routine monitoring and allow safety personnel to focus more effectively on identified risks.

OBJECTIVES

- ❖ To develop an AI-based construction safety monitoring platform using Artificial Intelligence, Computer Vision, and Deep Learning.
- ❖ To automatically detect construction workers from images and video using the YOLOv11 object detection model.
- ❖ To monitor Personal Protective Equipment (PPE) compliance by identifying safety helmets and safety vests.
- ❖ To detect PPE violations such as workers without helmets or safety vests.
- ❖ To enable continuous monitoring of construction-site activities through AI-powered image and video analysis.
- ❖ To identify unsafe conditions and potential safety risks using computer vision-based analysis.
- ❖ To provide safety alerts and violation information to support faster identification of safety issues.
- ❖ To calculate safety performance indicators such as compliance percentage, safety score, violation count, and risk level.
- ❖ To maintain safety records and historical data for monitoring and future analysis.
- ❖ To develop an interactive Streamlit dashboard for displaying safety monitoring results, analytics, and reports.
- ❖ To reduce dependence on manual safety inspection by automating routine PPE monitoring and violation detection.
- ❖ To support data-driven decision-making by providing meaningful safety insights and visual analytics.
- ❖ To improve worker safety and PPE compliance by enabling faster identification and response to safety violations.
- ❖ To demonstrate the practical application of AI technology in solving real-world construction industry safety.

WORK PROGRESS

- **Problem Identification & Requirement Analysis**
  Identified major construction-site safety challenges, especially PPE non-compliance and limitations of manual monitoring.

- **Technology & Architecture Selection**
  Selected Python, YOLOv11, Computer Vision, Roboflow, OpenCV, Streamlit, and Google Colab for developing the platform.

- **Dataset Collection & Preparation**
  Collected construction-site images and prepared datasets containing workers and PPE-related categories.

- **Dataset Annotation**
  Annotated images using Roboflow to identify workers, helmets, safety vests, and relevant safety conditions.

- **YOLOv11 Model Training**
  Trained and developed the object detection model using Google Colab and NVIDIA

- **GPU acceleration**.

- **PPE Detection Implementation**
  Integrated the trained model to detect workers and monitor helmet and safety-vest compliance.

- **Image & Video Analysis**
  Implemented OpenCV to process construction-site images and video frames for AI-based safety analysis.

- **Safety Violation Detection**
  Developed functionality to identify and record PPE violations and other detected safety issues.

- **Safety Analytics Development**
  Implemented analysis for compliance percentage, safety score, violation count, risk level, and safety grade.

- **Dashboard Development**
  Developed an interactive Streamlit dashboard for displaying monitoring results, analytics, safety intelligence, and reports.

- **Report & Data Management**
  Integrated data processing and visualization using Pandas and Matplotlib for safety records and reporting.

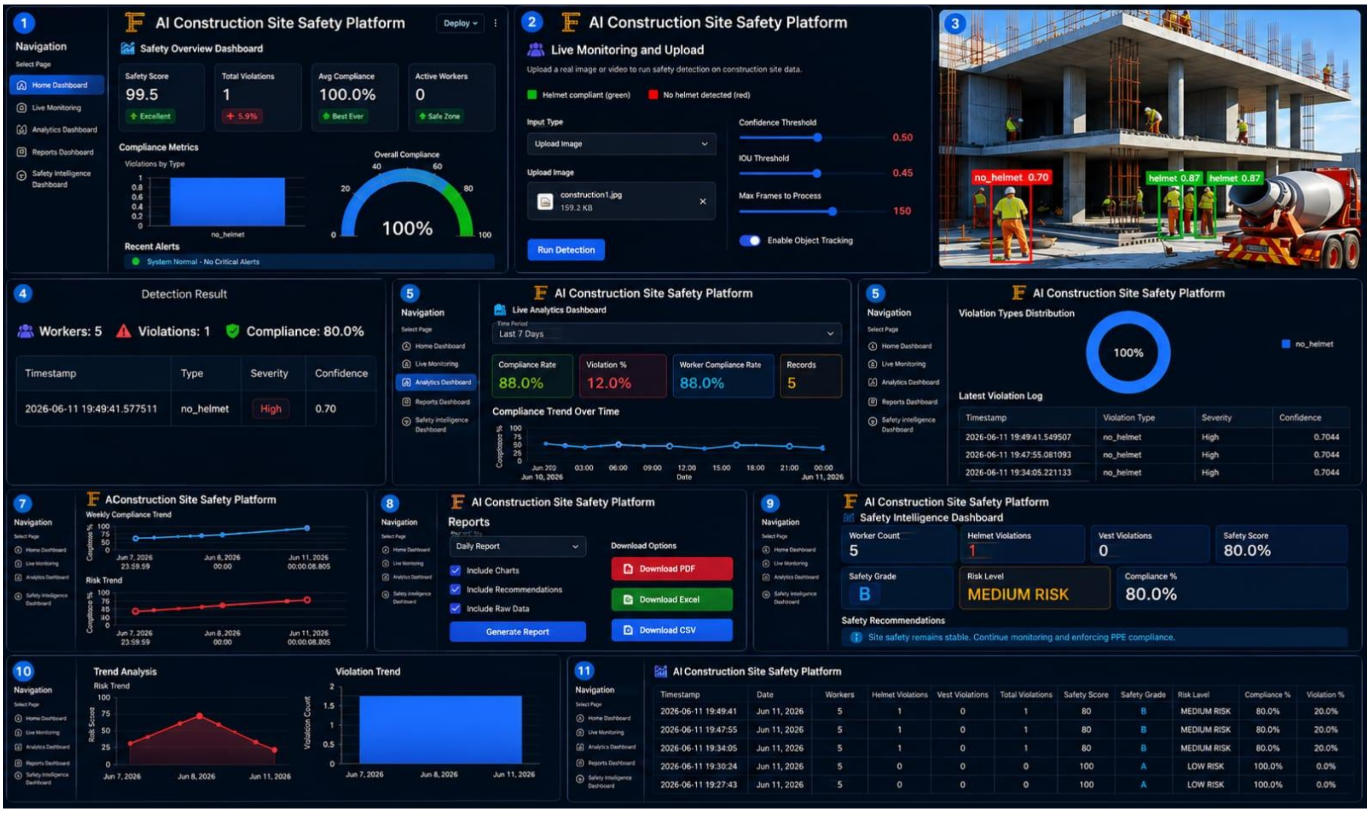
- **Testing & Validation**
  Tested the system with construction-site images and video inputs to evaluate worker detection, PPE monitoring, and safety analysis.

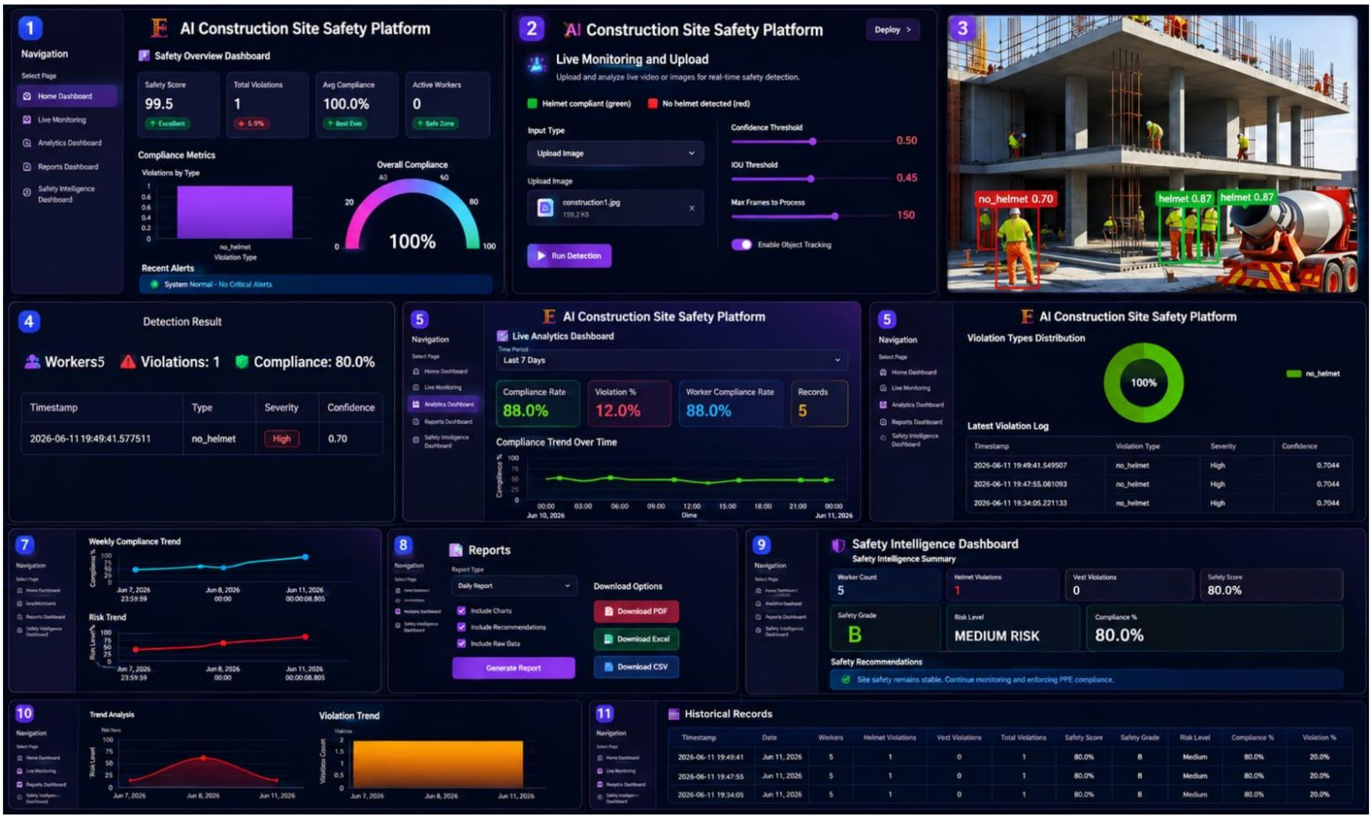
- **Current Progress**
  The major components of APEX PORTAL – Construction Safety, including AI-based detection, PPE monitoring, safety analysis, dashboard visualization, and reporting, have been successfully developed and integrated.

RESULTS

- ❖ The developed **APEX PORTAL – Construction Safety** platform successfully demonstrates AI-based monitoring of construction-site safety using **YOLOv11, Computer Vision, and Deep Learning**.
- ❖ Successfully developed an **AI-powered construction-site safety monitoring system**.
- ❖ Successfully detected **workers** from construction-site images and video.
- ❖ Implemented **helmet detection** for monitoring worker PPE compliance.
- ❖ Implemented **safety-vest detection** to identify whether required protective equipment is being used.
- ❖ Successfully identified **PPE violations** such as missing helmets and safety vests.
- ❖ Displayed **detection results and confidence information** for identified objects.
- ❖ Implemented **AI-powered image and video analysis** for construction safety monitoring.
- ❖ Developed **real-time/near-real-time safety monitoring capabilities** for supported video inputs.
- ❖ Generated **safety violation records** for further analysis.
- ❖ Implemented **PPE compliance percentage** calculation.
- ❖ Implemented **safety score** calculation based on detected safety conditions.
- ❖ Added **risk-level assessment** to provide a clearer understanding of site safety conditions.
- ❖ Implemented **violation count and safety statistics** for monitoring overall performance.
- ❖ Developed **historical safety records** for reviewing previously detected violations.
- ❖ Created an interactive **Streamlit dashboard** for centralized safety monitoring.
- ❖ Added **analytics and visualizations** to make safety information easier to understand.
- ❖ Integrated **safety reports and data handling** for documentation and analysis.

- ❖ Provided a structured platform for **safety intelligence and decision support**.
- ❖ Demonstrated that AI can assist in reducing the workload associated with continuous manual PPE inspection.
- ❖ Demonstrated the practical application of **AI and Computer Vision in the construction industry**.


AI Construction Site Safety Platform
Navigation
Home Dashboard
Live Monitoring
Analytics Dashboard
Reports Dashboard
Safety Intelligence Dashboard
Safety Overview Dashboard
Safety Score
99.5
Total Violations
1
Avg Compliance
100.0%
Active Workers
0
Compliance Metrics
Overall Compliance
100%
Recent Alerts
Live Monitoring and Upload
Helmet compliant (green)
No helmet detected (red)
Input Type
Upload Image
Confidence Threshold
0.50
IOU Threshold
0.45
Max Frames to Process
150
Enable Object Tracking
Run Detection
no_helmet 0.70
helmet 0.87
Detection Result
Workers: 5
Violations: 1
Compliance: 80.0%
Timestamp
Type
Severity
Confidence
2026-06-11 19:49:41.577511
no_helmet
High
0.70
Live Analytics Dashboard
Compliance Rate
88.0%
Violation %
12.0%
Worker Compliance Rate
88.0%
Records
5
Compliance Trend Over Time
Violation Types Distribution
100%
Latest Violation Log
Weekly Compliance Trend
Risk Trend
Reports
Daily Report
Download Options
Include Charts
Include Recommendations
Include Raw Data
Generate Report
Download PDF
Download Excel
Download CSV
Safety Intelligence Dashboard
Worker Count
Helmet Violations
Vest Violations
Safety Score
80.0%
Safety Grade
B
Risk Level
MEDIUM RISK
Compliance %
80.0%
Safety Recommendations
Site safety remains stable. Continue monitoring and enforcing PPE compliance.
Trend Analysis
Violation Trend


AI Construction Site Safety Platform
Navigation
Safety Overview Dashboard
Safety Score
99.5
Total Violations
1
Avg Compliance
100.0%
Active Workers
0
Compliance Metrics
Overall Compliance
100%
Recent Alerts
Live Monitoring and Upload
Upload and analyze live video or images for real-time safety detection.
Helmet compliant (green)
No helmet detected (red)
Confidence Threshold
0.50
IOU Threshold
0.45
Max Frames to Process
150
Run Detection
no_helmet 0.70
helmet 0.87
Detection Result
Workers5
Violations: 1
Compliance: 80.0%
Timestamp
Type
Severity
Confidence
2026-06-11 19:49:41.577511
no_helmet
High
0.70
Live Analytics Dashboard
Compliance Rate
88.0%
Violation %
12.0%
Worker Compliance Rate
88.0%
Records
5
Compliance Trend Over Time
Violation Types Distribution
100%
Latest Violation Log
Weekly Compliance Trend
Risk Trend
Reports
Download Options
Generate Report
Download PDF
Download Excel
Download CSV
Safety Intelligence Dashboard
Safety Intelligence Summary
Worker Count
5
Helmet Violations
1
Vest Violations
0
Safety Score
80.0%
Safety Grade
B
Risk Level
MEDIUM RISK
Compliance %
80.0%
Safety Recommendations
Site safety remains stable. Continue monitoring and enforcing PPE compliance.
Trend Analysis
Violation Trend
Historical Records

CONCLUSION

The **APEX PORTAL – Construction Safety** project successfully demonstrates how modern **Artificial Intelligence (AI), Computer Vision, and Deep Learning** technologies can be applied to address real-world safety challenges in the construction industry. The platform provides an automated approach to monitoring workers and identifying PPE-related safety violations.

The implementation of **YOLOv11** enables the system to detect workers and safety equipment such as **helmets and safety vests** from construction-site images and video. This helps identify cases where workers may not be following required PPE practices and provides useful information for further safety analysis.

The platform also converts detection results into meaningful safety information, including **PPE compliance percentage, safety scores, violation counts, risk levels, and safety records**. These features help provide a better understanding of the overall safety condition of a construction site and can support safety teams in identifying areas that require attention.

The **Streamlit-based dashboard** provides a centralized interface for viewing detection results, safety analytics, historical records, and reports. This makes the system more user-friendly and allows safety information to be presented in a clear and organized manner.

One of the major benefits of the project is the reduction of dependence on **continuous manual PPE inspection**. Instead of relying only on human observation, AI-based monitoring can assist safety personnel by automatically analyzing visual information and highlighting potential violations. This can help save monitoring time and improve the speed at which safety issues are identified.

The project also provides valuable practical experience in **AI model development, dataset preparation, object detection, image and video processing, data analysis, dashboard development, and deployment**. The integration of technologies such as **Python, YOLOv11, Roboflow, OpenCV, Google Colab, Pandas, Matplotlib, and Streamlit** demonstrates an end-to-end approach to developing an industry-oriented AI solution.

Overall, **APEX PORTAL – Construction Safety** provides an intelligent and data-driven approach to construction safety management. The platform can contribute to **better PPE compliance, improved safety awareness, faster identification of safety violations, and more effective safety decision-making**.

REFERENCES

APEX PORTAL – Construction Safety

1. **Ultralytics YOLO Documentation** – YOLOv11 object detection, training, and model implementation.
2. **Roboflow Documentation** – Dataset collection, image annotation, preprocessing, and computer vision dataset management.
3. **OpenCV Documentation** – Image processing, video processing, and computer vision operations.
4. **Python Documentation** – Python programming concepts and libraries used for AI-based application development.
5. **PyTorch Documentation** – Deep learning framework supporting model training and inference.
6. **Google Colab Documentation** – Cloud-based environment used for AI model development and GPU-accelerated training.
7. **Streamlit Documentation** – Development of interactive dashboards and data-driven web applications.
8. **Pandas Documentation** – Data processing, analysis, and management of safety records.
9. **Matplotlib Documentation** – Data visualization and graphical representation of safety analytics.
10. **NumPy Documentation** – Numerical computation and array-based data processing.
11. **ngrok / Pyngrok Documentation** – Secure tunneling and online access for locally hosted applications.
12. **Occupational Safety and Health Administration (OSHA)** – Construction safety practices and Personal Protective Equipment (PPE) guidelines.
13. **National Institute for Occupational Safety and Health (NIOSH)** – Workplace safety research and construction safety guidelines.

14. **Construction Industry Safety Guidelines** – General principles and recommended practices for PPE compliance and workplace hazard prevention.